

Phase 1: Mine Licensing & Mine GIS Management System (Mine ERP)

Project Overview

This proposal outlines the first phase of the **Mine Licensing & Mine GIS Management System (Mine ERP)** for the **National Gem & Jewellery Authority (NGJA)**. The system digitizes the complete mining license lifecycle, enabling miners to apply online, officers to manage approvals electronically, and management to monitor the entire licensing process. Following license approval, the system integrates with the Mine GIS platform to register mine locations, GPS coordinates, photographs, and inspection data, providing a centralized foundation for efficient mine management and future monitoring.

Phase 1 Scope

1. Miner Registration & Account Management

Prospective miners will be able to register securely through the Mine ERP platform before applying for a mining license.

The system will support:

- Secure Miner Registration
- User authentication
- Profile management
- Password management
- Document management

Registration information may include:

- National Identity Card (NIC)
- Tax Identification Number (TIN)
- Land Deed
- Affidavits
- Ownership documents
- Other legally required supporting documents

Only registered users will be permitted to submit mining license applications.

2. Digital Mining License Application

Registered miners will be able to submit mining license applications electronically without visiting regional offices.

The application process will allow applicants to:

- Create new mining license applications
- Enter mine and land information
- Upload required legal documents
- Review submitted information

- Submit applications electronically
- Track application status

Each application will receive a unique application reference number for tracking throughout the approval process.

3. Regional Office Review & Initial Site Inspection

After submission, applications will be automatically assigned to the relevant Regional Office based on the mine location.

Regional officers will be responsible for:

- Reviewing submitted documentation
- Verifying ownership documents
- Conducting the initial site inspection
- Recording inspection observations
- Verifying applicant information
- Recommending approval or requesting corrections

If deficiencies are identified, the application may be returned to the applicant for correction before continuing the approval process.

4. Public Notice & Mandatory Waiting Period

Once the Regional Officer recommends approval, the application will enter the statutory public notification stage.

The system will manage:

- Automatic initiation of the mandatory 14-day waiting period
- Waiting period countdown
- Application status monitoring
- Objection recording
- Hold management

If objections are received during the waiting period:

- The application status will automatically change to On Hold
- Officers will be notified
- Further approval will be suspended until objections are resolved

If no objections are received, the application will automatically proceed to the next approval stage.

5. Legal Division Review

Applications that successfully complete the waiting period will be forwarded electronically to the NGJA Legal Division.

The Legal Division will be able to:

- Review all submitted documentation
- Verify legal compliance
- Record legal observations
- Request additional information
- Approve or reject applications

Only legally approved applications will proceed to the Chairman for final approval.

6. Chairman Approval

The final approval stage will be managed digitally through the Chairman's approval portal.

The Chairman will be able to:

- Review complete application details
- Review inspection recommendations
- Review Legal Division recommendations
- Approve or reject mining license applications
- Record approval remarks

Upon approval, the system will automatically issue the mining license and update the application status.

7. Mining License Issuance

After final approval, the system will generate the official mining license.

The system will support:

- Digital mining license generation
- License numbering
- License validity management
- License history
- Downloadable license documents
- License verification

The approved license will become the official record for the registered mine.

8. Mine Registration into the Mine GIS System

Once the mining license has been issued, the Regional Officer will conduct a second field visit to officially register the mine within the Mine GIS Management System.

The officer will collect:

- Mine GPS coordinates
- Latitude
- Longitude
- Mine location
- Mine boundary
- Mine photographs
- General mine information

Each registered mine will receive a unique Mine GIS record linked directly to its approved mining license.

This establishes the foundation for digital mine monitoring and future inspections.

9. Mine GIS & Inspection Management

After the mine has been registered within the Mine GIS system, the Mine ERP will support ongoing field inspection and monitoring activities.

The system will allow authorized officers to:

- Record mine inspections
- Capture GPS locations
- Update mine boundaries
- Upload inspection photographs
- Record environmental observations
- Record operational conditions
- Maintain inspection history

Each inspection will be permanently linked to the corresponding licensed mine.

10. Management Dashboard & Monitoring

Management personnel will have access to a centralized dashboard for monitoring the complete mining license lifecycle and Mine GIS information.

The dashboard will provide visibility into:

New license applications

- Applications awaiting approval
- Applications under legal review
- Applications in the 14-day waiting period
- Objections received
- Approved mining licenses
- Mine GIS locations
- Registered mines
- Inspection history

- Regional officer activities
- Mine operational status

This enables informed operational planning and strategic decision-making.

Security Considerations

Initial Approach

The system will authenticate all users before granting access to licensing and Mine GIS functions. All application data and supporting documents will be securely transmitted and stored using encrypted communication channels.

Different access levels will be provided for:

- Miners
- Regional Officers
- Legal Officers
- Chairman
- System Administrators

Recommended Approach

For production deployment, additional security measures are recommended, including:

- Multi-factor authentication (MFA)
- Role-based access control
- Secure document storage
- Encrypted API communication
- Digital audit trails
- Activity logging
- Document integrity verification
- Secure backup and disaster recovery

These measures will strengthen operational security, protect sensitive legal documents, and improve accountability throughout the licensing process.

Process Study Requirement

Prior to implementation, it is recommended that the complete mining license approval workflow be reviewed in detail.

This review should include:

- Current mining license application procedures
- Regional office workflows
- Site inspection procedures
- Public notification and objection handling

- Legal Division review processes
- Chairman approval workflow
- Mining license issuance procedures
- Mine registration procedures
- Existing Mine GIS data collection practices

Understanding these operational processes will ensure that the digital platform accurately reflects statutory requirements while improving efficiency, transparency, and traceability.

Expected Deliverables (Phase 1)

- Secure Mine Licensing Management Portal
- Miner registration and profile management
- Digital mining license application workflow
- Supporting document management
- Regional Office review and inspection management
- Automated 14-day waiting period management
- Objection handling workflow
- Legal Division approval workflow
- Chairman approval workflow
- Digital mining license issuance
- Mine registration within the Mine GIS platform
- GPS location and mine boundary recording
- Mine photograph management
- Basic Mine GIS dashboard
- Inspection history management
- Management reporting and monitoring dashboard

Future Enhancements

- Mobile inspection application for field officers
- QR code identification for licensed mines
- Digital signatures for approvals
- Automated inspection scheduling
- Renewal and expiration management
- SMS and email notifications
- Drone image integration
- Satellite imagery integration
- Environmental monitoring integration
- GIS-based mine analytics and heat maps
- Integration with the complete NGJA Enterprise ERP platform

Conclusion

The **Mine Licensing & Mine GIS Management System (Mine ERP)** provides a centralized digital platform that streamlines the complete mining license approval process, from miner registration and application submission to regional inspections, legal review, and final approval. Following license issuance, the system integrates with the Mine GIS platform to register and monitor mine locations, boundaries, and inspection data. By combining licensing and GIS management into a single solution, the platform enhances efficiency, transparency, regulatory compliance, and informed decision-making across the National Gem & Jewelry Authority.